NETWORK CABLES AND RJ-45 TERMINATION

Question 1 — Cat6 cable diagram and T568B pin order

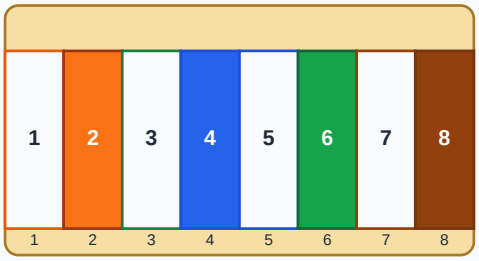
Cat6 is a twisted-pair Ethernet cable used for wired local-area networks. The T568B standard specifies the order of the eight conductors at the RJ-45-style 8P8C plug. For a normal straight-through cable, use T568B at both ends.

Cat6 Ethernet Cable — T568B Wiring Order

Pin order, left to right:

- Pin 1: White/Orange
- Pin 2: Orange
- Pin 3: White/Green
- Pin 4: Blue
- Pin 5: White/Blue
- Pin 6: Green
- Pin 7: White/Brown
- Pin 8: Brown

RJ-45 plug (front view)



Hold the plug with gold contacts facing you and the clip on the back/underside; confirm the viewing orientation before wiring.

T568B color order: Pin 1 White/Orange, Pin 2 Orange, Pin 3 White/Green, Pin 4 Blue, Pin 5 White/Blue, Pin 6 Green, Pin 7 White/Brown, Pin 8 Brown.

Important: The plug must be viewed in the same orientation as the diagram; flipping the plug changes left-to-right order. Cat6 cable uses four twisted pairs. Keep the twists close to the connector to help maintain signal quality.

Question 2 — Three network cable types and their uses

Cable type	Example device / use	Why it fits
Cat5e twisted-pair Ethernet	Wi-Fi router to a home desktop or a café POS terminal	Cat5e supports Gigabit Ethernet up to 100 metres in a compliant installation, making it a cost-effective choice for everyday office, home, and point-of-sale connections.
Cat6 twisted-pair Ethernet	Gaming console or smart TV connected to a gigabit switch/router	Cat6 offers better crosstalk performance and can support 1 Gb/s to 100 metres; it can support 10 Gb/s over shorter distances depending on installation conditions.
Fiber-optic cable	Link between network switches in a college, event venue, or IPL streaming facility	Fiber carries data as light, supports high bandwidth and long distances, and is immune to electromagnetic interference. It needs compatible optical ports/transceivers.

Other useful connector types

RJ-45/8P8C: Commonly used to terminate copper Ethernet cables and plugs into a router, switch, PC, console, or network printer. **LC or SC fiber connectors:** Commonly used with fiber-optic links and compatible switch transceivers. The correct connector depends on the cable and the network equipment ports.

A Wi-Fi router may still use Ethernet cables for its wired internet connection and for connecting local devices. Wi-Fi is wireless; Cat5e/Cat6 are the physical wired links.

Question 3 — RJ-45 crimping process and common mistakes

A typical crimping tutorial demonstrates how to prepare a twisted-pair cable, arrange the conductors in the T568B order, insert them into a modular plug, and crimp the plug. The steps below are written in my own words as a practical summary. No specific video was verified or watched for this document, so use these steps alongside a reputable crimping demonstration.

Step 1: Measure and cut. Measure the required cable length and cut it cleanly with a cable cutter.

Step 2: Strip the jacket. Use a cable stripper to remove a short section of the outer jacket without nicking the inner wires.

Step 3: Separate the pairs. Untwist and straighten the eight conductors only as much as needed. Keep twists close to the plug.

Step 4: Arrange T568B. Place the wires in this order from pin 1 to pin 8: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

Step 5: Trim evenly. Hold the wires in order and cut their ends straight so all conductors are the same length.

Step 6: Insert into the plug. Slide the wires fully into the RJ-45 plug, checking that each wire reaches its correct channel and the cable jacket sits under the plug's strain-relief tab.

Step 7: Crimp. Place the plug in a compatible crimping tool and squeeze firmly to seat the contacts and strain relief.

Step 8: Repeat and test. For a straight-through cable, terminate the other end with the same T568B order. Use a cable tester to verify all eight conductors are connected in the correct sequence.

Two common mistakes to avoid

Wrong wire order: Mixing up the green and orange pairs or reversing the sequence can cause a failed cable test or unreliable connection. Check the T568B order before crimping. **Wires or jacket not fully inserted:** If a conductor does not reach the end of its channel, or the outer jacket is not secured under the strain relief, the cable may fail or break with movement.

Safety and quality tip: Use a crimp tool and cable tester designed for the plug and cable type. Avoid cutting too deeply into the conductors, and do not rely on link lights alone when a tester is available.

Question 4 — Cabling a gaming zone for an IPL screening event

Recommendation: Use Cat6 copper Ethernet with RJ-45/8P8C connectors for each nearby console, smart TV, streaming PC, and wired access point. Connect these cables to a gigabit or faster Ethernet switch, then connect the switch to the router/internet gateway. For long runs between distant rooms or network cabinets, use fiber-optic cable with compatible optical transceivers.

Why these choices work

Cat6 provides a reliable wired connection with low interference when installed correctly, which is useful for multiplayer gaming, streaming, and stable screen feeds. A switch gives each connected device its own network port and avoids depending on wireless signal strength for fixed equipment. Fiber is suitable for longer backbone links because it supports high bandwidth and is not affected by electromagnetic interference from power cables.

Connector and setup choices

• Use RJ-45/8P8C plugs or, preferably for permanent runs, Cat6 keystone jacks and patch panels at the copper cable ends.

• Use a managed or unmanaged gigabit switch with enough ports for consoles, TVs, streaming PCs, router uplink, and spare capacity.

• Use HDMI from each console or streaming device to its screen; Ethernet carries network data, not the video signal from the console to the display.

• Use fiber connectors such as LC/SC only where the selected fiber cable and switch transceivers support them.

• Label both ends of each cable, keep copper Ethernet runs within the standard 100-metre channel limit, and test each run before the event.

Planning note: The appropriate HDMI version and display cable depend on the resolution and refresh rate required. Ethernet connects devices to the network; HDMI or DisplayPort carries video from a console/PC to a screen.

Question 5 — Photo example and description

Example photo reference: a Cat6 Ethernet patch cable with RJ-45 connectors plugged into a Wi-Fi router and a desktop computer or network switch. This is a representative example for the assignment, not a photo taken in your home or college.

Illustrative setup: Cat6 Ethernet cable with RJ-45 connectors

The cable links network equipment; end devices may connect to other switch ports.

Description: The blue cable shown represents a Cat6 twisted-pair Ethernet patch cable. Its clear modular plugs are RJ-45/8P8C connectors. One end can connect to a router LAN port and the other to a switch, desktop PC, POS terminal, or other wired network device. It carries network data and does not directly carry the video signal from a console to a display.

If your teacher requires an actual photo: Take a picture of a real cable behind your home router or at college, or insert a properly credited online photo. Then replace this illustrative example with your own photo and identify the cable category, connector, and connected devices.